

Rekrut AI – Product & Technical Bible

Version 2.0 • January 02, 2026

Audience: Investors, Engineering/ML teams, and prospective customers. This document defines what Rekrut AI is building, why it wins, and how it is implemented end-to-end.

1. Executive Summary

Rekrut AI is a SaaS platform that modernizes hiring with AI-first workflows. It combines skill validation, interview intelligence, candidate-recruiter collaboration, and analytics in one system so teams can hire faster, more fairly, and with higher confidence. The platform supports candidates (practice, guidance, transparency) and recruiters (pipeline automation, ranking, insights). The initial feature scope includes AI skill assessments, AI interview coaching, AI profile matching, document verification, hiring dashboards, automated scheduling, collaboration tools, and onboarding support.

2. Problem, Users, and Jobs-to-be-Done

- Recruiters spend significant time on repetitive tasks: screening, scheduling, chasing feedback, and compiling notes.
- Candidates face opaque processes, inconsistent evaluations, and limited feedback; drop-off is common.
- Hiring managers struggle to compare candidates objectively across interviewers and stages.

Primary users: Candidates, Recruiters, Hiring Managers, and Admin/HR Ops.

3. Product Overview and Modules

The platform is organized into modules. The feature set below is based on your Rekrut AI scope document.

AI Skill Assessments

- Generate role-specific tests (coding, data analysis, algorithm design, domain quizzes).
- Auto-grade and provide feedback and improvement suggestions to candidates.
- Anti-cheating controls (proctoring options, plagiarism checks, AI-usage policy).

AI Interview Coaching (Candidate)

- AI coach simulates interviews with tailored questions by role and seniority.
- Provides actionable feedback on structure, clarity, confidence, and delivery.

AI Interview Intelligence (Recruiter)

- Interview capture (notes/transcripts) and AI summaries for faster review.
- Competency tagging, structured rubric scoring, and highlight reels for key moments.

AI Profile Matching & Candidate Ranking

- Match resumes/portfolios/assessment outcomes to job requirements.
- Rank candidates with explainable scoring (skills, experience, performance, potential).

Document Management & Verification

- Secure document upload (diplomas, certificates, IDs where relevant).
- Extraction (OCR/NLP), verification workflows, and audit trails.

Hiring Dashboard & Workflow

- Pipeline stages configurable per role (screen, assessment, interviews, offer).
- Team collaboration: notes, tags, mentions, decision logs.

Automated Interview Scheduling

- Availability collection and automatic slot selection; calendar invites and reminders.
- Support panel interviews, time zones, rescheduling, and buffer rules.

Performance Analytics

- Candidate-level analytics (assessment performance, interview rubric trends).
- Process analytics (time-to-hire, stage conversion, interviewer calibration).

Compliance, Privacy & Security

- Consent, notices, and audit reporting for automated decision tools.
- Role-based access, encryption, retention policies, and privacy requests.

Job Recommendations & Onboarding Support

- Recommend jobs to candidates based on profile signals and preferences.
- Onboarding checklist + AI guidance for paperwork and policy acknowledgments.

Source: Rekrut AI scope/features list provided by you. (Internal reference: OmniHire.docx)

4. Technical Architecture

Rekrut AI is designed as a cloud-native, modular platform with separately scalable core services and AI services.

Rekrut AI – High-Level Reference Architecture

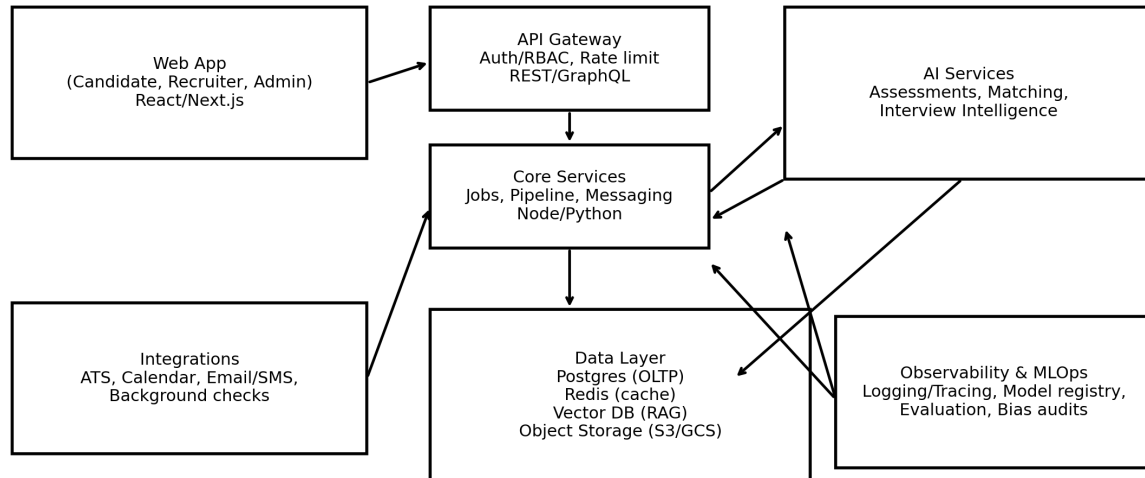


Figure 1. Reference architecture: frontends → API gateway → core services ↔ AI services → shared data layer, with integrations and observability/MLOps.

4.1 Core Components

- **Frontend:** Candidate Portal, Recruiter Dashboard, Admin Console (React/Next.js).
- **API Gateway:** Authentication (OAuth/JWT), RBAC, rate limiting, request validation, API versioning.
- **Core Services:** Jobs, Applications/Pipeline, Messaging/Notifications, Scheduling, Billing/Plans, Organization settings.
- **AI Services:** Assessment generation and grading, resume parsing & matching, interview transcription & summarization, feedback generation, recommendation engine.
- **Data Layer:** PostgreSQL (OLTP), Redis (caching/queues), Object storage (documents/media), Vector DB (semantic search/RAG), Analytics warehouse (optional).
- **Integrations:** ATS, calendar (Google/Microsoft), email (SendGrid), SMS (Twilio), background checks, identity verification.

4.2 Full Infrastructure (End-to-End)

Cloud & Networking

- Multi-environment setup: dev/stage/prod with isolated VPC/VNet.

- Public edge: CDN + WAF + TLS termination; API gateway behind load balancer.
- Private subnets for databases, caches, and internal services; egress controls + NAT.
- Secrets management (KMS + secret store); zero-trust service-to-service auth (mTLS).

Compute

- Containerized services (Docker) orchestrated by Kubernetes or managed container platform.
- Separate node pools for CPU workloads (web, core services) and GPU workloads (model inference/training).
- Autoscaling based on latency/queue depth; rolling deploys with health checks.

Data & Storage

- PostgreSQL with read replicas and PITR backups; schema migrations with CI/CD gates.
- Redis for caching sessions, rate limits, and background jobs.
- Object storage for resumes, certificates, interview recordings; pre-signed URLs; lifecycle policies.
- Optional warehouse (BigQuery/Snowflake/Redshift) for long-term analytics and cohort reporting.

Observability

- Centralized logs (structured JSON), metrics, and distributed tracing (OpenTelemetry).
- Dashboards for API latency, queue depth, assessment throughput, interview processing time.
- Alerting (SLO-based) + incident runbooks.

Security

- RBAC/ABAC with least privilege; separate org tenancy controls.
- Encryption in transit (TLS) and at rest (KMS-managed keys).
- Vulnerability scanning (images/dependencies), SAST/DAST, regular pen testing.
- Audit logs for administrative actions and automated decision-making events.

MLOps

- Model registry, versioning, and feature store (optional).
- Evaluation harness: accuracy, drift, latency, fairness metrics; human review loops.
- Bias audit support and reporting for jurisdictions that require it (e.g., NYC AEDT law).
- Safe prompting + guardrails; red-team tests against prompt injection and data exfiltration.

CI/CD

- Pipeline: lint/test → build → scan → deploy → post-deploy verification.
- Blue/green or canary releases for high-risk changes; quick rollback.
- Infrastructure-as-code (Terraform/Pulumi) + policy-as-code.

5. Data Model (ERD Overview)

Below is a practical ERD-level blueprint for the first production release. It captures hiring workflows, assessments, interviews, documents, and messaging.

Rekrut AI – Core Data Model (ERD Overview)

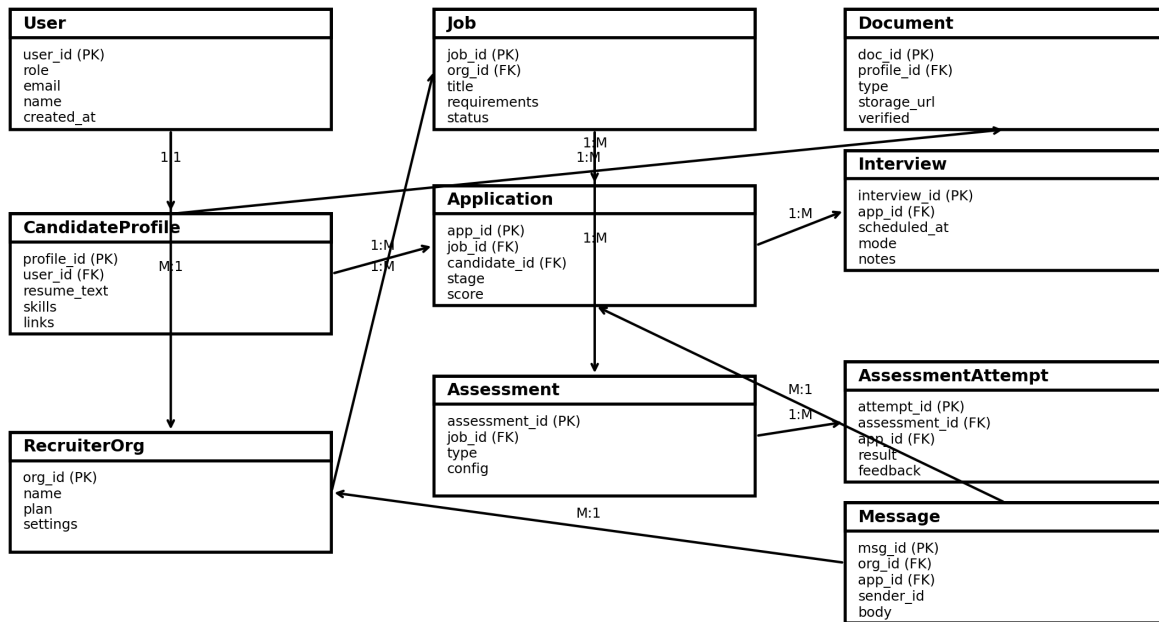


Figure 2. ERD overview (core entities and relationships).

6. Competitive Landscape

Rekrut AI competes with specialized point solutions. The strategy is to win by combining skill validation + interview intelligence + workflow automation + candidate experience in one platform.

HireVue	Video interviewing plus AI interview insights and assessments.	Broader end-to-end pipeline (matching, document verification, coaching, onboarding) with configurable modules.
Harver / Pymetrics	Behavioral / soft-skill assessments with audited AI; high-volume hiring.	Combine behavioral signals with hard skill validation, interview intelligence, richer collaboration, and unified analytics.
Codility	Best-in-class technical tests and live coding interviews for engineers.	Expand beyond engineering while keeping technical depth; unify with scheduling, ranking, and cross-role workflows.

Competitive source points used for this section come from vendor product pages and official statements (see References).

7. SWOT Analysis

Strengths

- All-in-one platform across screening → interview → decisioning → onboarding.
- Configurable workflows and collaboration built-in.
- AI modules can scale independently; strong data foundation for analytics.

Weaknesses

- Breadth increases product complexity and time-to-excellence in every module.
- High trust bar: fairness, explainability, and model governance must be strong from day one.

Opportunities

- Growing demand for skill-based hiring and AI-assisted recruiting.
- Tool consolidation: replace multiple point solutions with one integrated system.
- Enterprise compliance tooling (bias audit support, transparency reporting).

Threats

- Entrenched vendors and fast-moving startups.
- Regulatory changes in AI hiring; audits, notices, and data rights requirements.
- Candidate distrust if AI feels opaque or unfair.

8. Roadmap (12–24 Months)

Phase 1 (0–3 months): MVP

- Candidate + recruiter portals, organizations, jobs, applications, basic pipeline.
- Assessment v1 (question bank + auto-grading), scheduling v1, notifications.
- Resume parsing and matching v1 (rules + embeddings).

Phase 2 (4–9 months): Product-Market Fit

- Interview intelligence v1: transcripts, summaries, rubrics, team notes.
- Analytics v1: time-to-hire, stage conversion, assessment distributions.
- Security hardening: RBAC, audit logs, retention policies, incident playbooks.

Phase 3 (10–15 months): Enterprise Readiness

- Integrations: ATS (Greenhouse/Lever), calendars, SSO, webhooks.
- MLOps: model registry, eval harness, drift monitoring, fairness metrics.
- Compliance toolkit: notices, audit summaries, model cards, documentation exports.

Phase 4 (16–24 months): Scale + Differentiators

- Personalized candidate coaching improvements; job recommendations v2.
- Advanced anti-fraud for assessments; multilingual support.
- Marketplace integrations (background checks, ID verification, learning partners).

9. Compliance Notes (AI in Hiring)

AI used for employment decision support is increasingly regulated. New York City's AEDT framework requires (among other items) an independent bias audit within the prior 12 months, public posting of audit summaries, and advance notice to candidates before using an automated employment decision tool.

10. References

- Rekrut AI internal scope document (uploaded as OmniHire.docx) – feature list and initial architecture notes.
- HireVue – Product pages (video interviewing and Interview Insights): <https://www.hirevue.com/> and <https://www.hirevue.com/platform/interview-insights> (accessed 2026-01-02).
- Harver press release on acquiring Pymetrics: <https://harver.com/press/harver-acquires-pymetrics/> (accessed 2026-01-02).
- Codility – Platform overview and Interview (live coding): <https://www.codility.com/> and <https://www.codility.com/product-tour/interview/> (accessed 2026-01-02).
- NYC Department of Consumer and Worker Protection – Automated Employment Decision Tools guidance: <https://www.nyc.gov/site/dca/about/automated-employment-decision-tools.page> (accessed 2026-01-02).
- New York City AEDT rules: <https://rules.cityofnewyork.us/rule/automated-employment-decision-tools-2/> (accessed 2026-01-02).

- New York State Comptroller audit summary referencing Local Law 144 requirements (Dec 2, 2025): <https://www.osc.ny.gov/state-agencies/audits/2025/12/02/enforcement-local-law-144-automated-employment-decision-tools> (accessed 2026-01-02).